

Robotics Control Systems & Dynamics

A Comprehensive Thesis-Level Study on Modeling, Analysis, and Control

Department of Mechatronics & Robotics Engineering

Academic Year 2024/2025

Robotic System Architecture

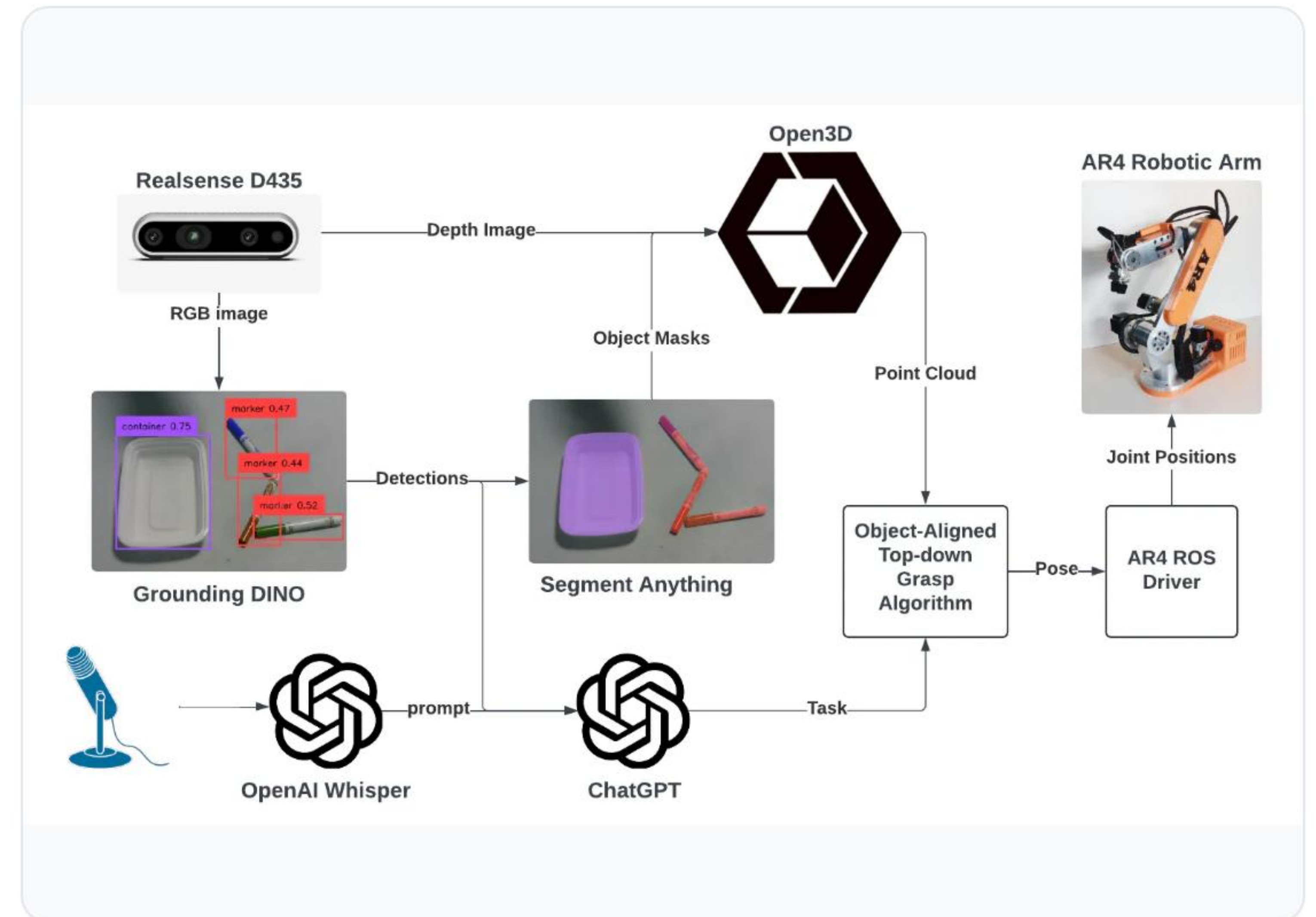
Hierarchy of Control

Modern robotics utilizes a multi-layered control architecture to bridge high-level cognition with low-level hardware execution.

 **Task Level:** Path planning and logic.

 **Control Level:** Motion profiles and PID loops.

 **Hardware Level:** Sensors, actuators, and signal conditioning.



| Hardware Interface: Sensors & Actuators

Category	Device Type	Key Parameters	Application in Control
Actuators	BLDC Motors / Steppers	Torque, RPM, Power Density	Joint position and velocity control
Exteroceptive	LiDAR / Stereo Vision	FOV, Resolution, Frame Rate	SLAM and Obstacle Avoidance
Proprioceptive	IMUs / Encoders	Sampling Rate, Drift, CPR	State estimation and feedback
Force	Strain Gauges / Piezo	Sensitivity, Bandwidth	Impedance and Haptic feedback

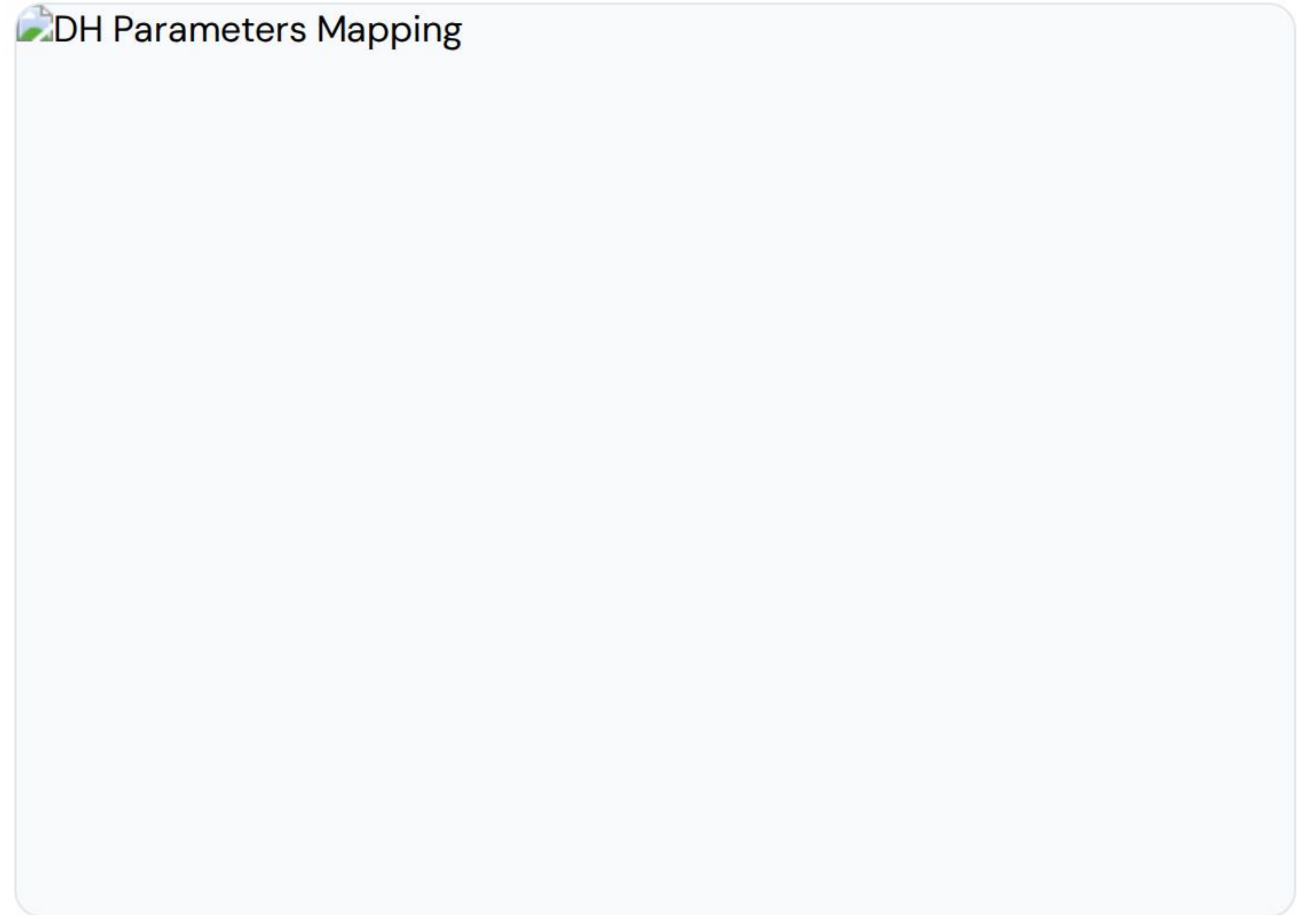
Kinematic Modeling: DH Parameters

Denavit–Hartenberg (DH) convention provides a systematic way to assign coordinate frames to robot links, defining the transformation between adjacent frames.

$${}^{i-1}T^i = \text{Rot}(z, \theta_i) \text{Trans}(z, d_i) \text{Trans}(x, a_i) \text{Rot}(x, \alpha_i)$$

Parameters: θ (Joint angle), d (Link offset), a (Link length), α (Link twist).

 DH Parameters Mapping



| Forward and Inverse Kinematics

Forward Kinematics

Maps joint coordinates q to end-effector pose X . Unique solution exists for open chains.

$$X = f(q)$$

Inverse Kinematics

Maps desired pose X back to joint space. Often non-unique or has no solution (singularities).

$$q = f^{-1}(X)$$

Jacobian Analysis

Relates joint velocities to end-effector velocity. Critical for force mapping.

$$\dot{X} = J(q) \dot{q}$$

| Dynamics: Euler-Lagrange Formulation

Dynamic modeling is essential for high-speed motion and force control. The Euler-Lagrange approach uses energy scalar functions.

$$M(q)\ddot{q} + C(q, \dot{q})\dot{q} + G(q) = \tau$$

 **M(q)**: Inertia Matrix (Positive Definite).

 **C(q, q')**: Coriolis and Centripetal forces.

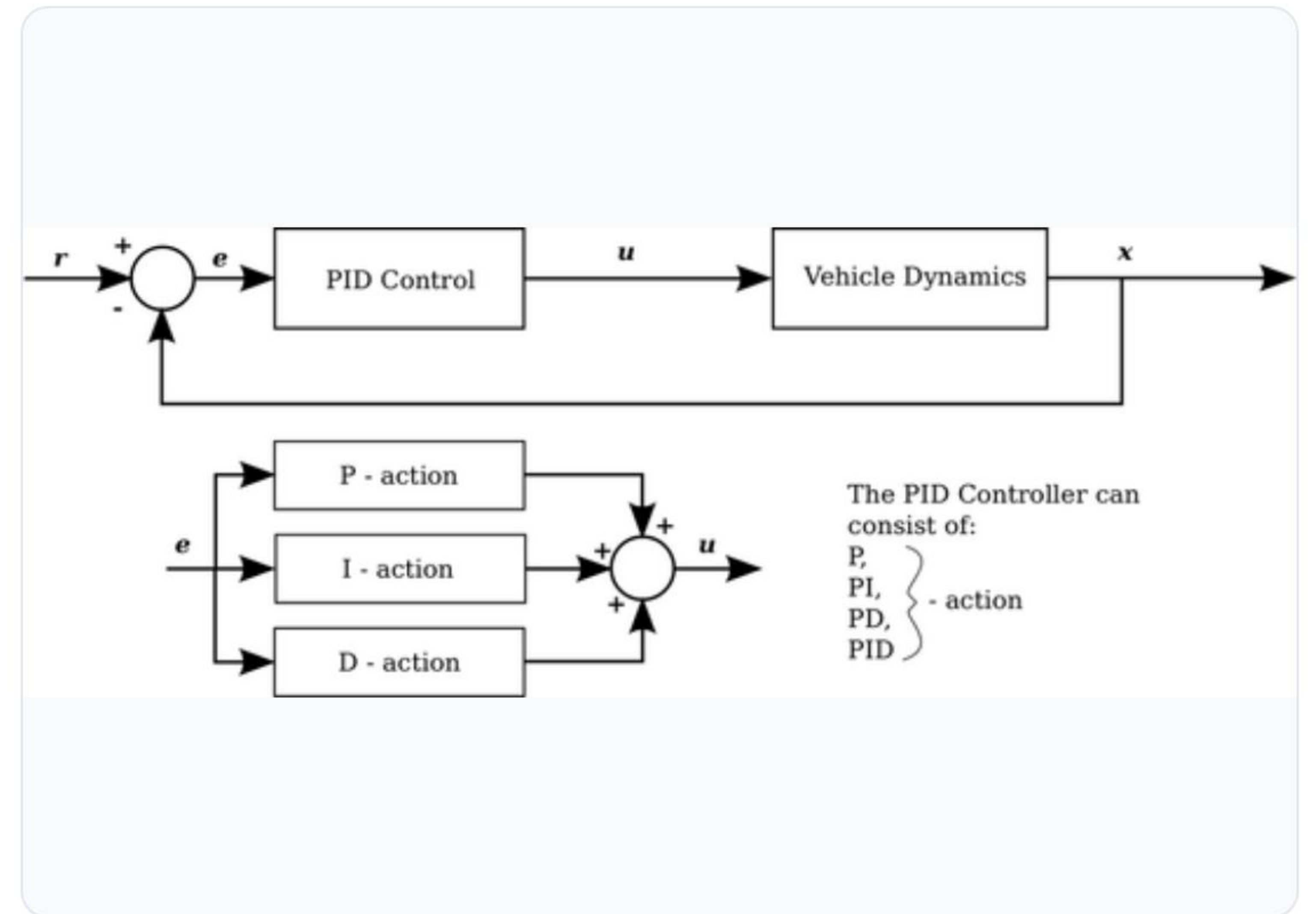
 **G(q)**: Gravitational loading.

Classical Control: PID Regulation

The Proportional-Integral-Derivative (PID) controller remains the industry standard for joint-level regulation due to its simplicity.

$$u(t) = K_p e(t) + K_i \int e(\tau) d\tau + K_d \frac{de}{dt}$$

Limitations: Ignores multi-variable coupling and nonlinear dynamics (gravity, friction).



| Modern Control: State-Space & Stability

State Representation

Modeling high-order systems as a set of coupled first-order differential equations.

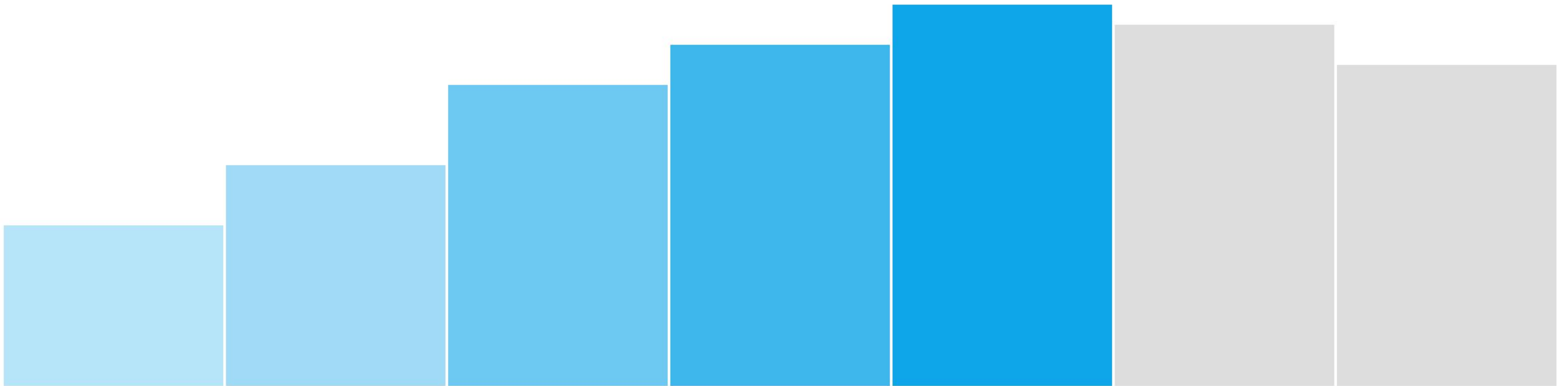
$$\dot{x} = Ax + Bu$$

Lyapunov Stability

Stability is proven by finding a positive-definite energy-like function $V(x)$ such that its derivative is negative semi-definite.

- 🛡️ Global Asymptotic Stability (GAS).
- ✅ Linear Quadratic Regulator (LQR) for optimal gain selection.

| Advanced Control: Model Predictive Control



MPC Features: Solves a constrained optimization problem at each time step over a prediction horizon. Ideal for handling actuators constraints and obstacle avoidance in real-time.

| Nonlinear Control Techniques

Feedback Linearization

Canceling nonlinear terms using the dynamic model to transform the system into a controllable linear form.

Sliding Mode Control

Driving the system state to a "sliding surface" and maintaining it there, providing high robustness to perturbations.

Backstepping

A recursive Lyapunov-based design for systems where the control input is separated from the nonlinearity by integrators.

| Handling Uncertainties

Method	Core Concept	Dealing with Parameters	Robustness
Adaptive Control	Online Parameter Estimation	Updates model gains in real-time	Moderate
Robust Control	Worst-case Design (H-infinity)	Fixed gains based on uncertainty bounds	High
Neural Control	Universal Approximation	Learns unknown dynamics via data	Variable

Real-World Applications

From Industrial Shop Floors to Humanoid Locomotion

Industrial Manipulators

Six-DOF (Degrees of Freedom) robots are the backbone of modern manufacturing. Control focuses on:

- 🎯 High-precision trajectory tracking.
- 🖐️ Singularity avoidance.
- 🛡️ Collaborative safety (Cobots).



| Mobile Robotics: Differential Drive

Non-holonomic Constraints

Mobile robots cannot move instantly in any direction. They follow rolling-without-slipping constraints.

$$\dot{x} = v \cos \theta, \dot{y} = v \sin \theta$$

Navigation Stack

Integration of AMCL (localization), TEB Local Planner (trajectory), and Costmaps for autonomous transit.

| Aerial Robotics: UAV Dynamics

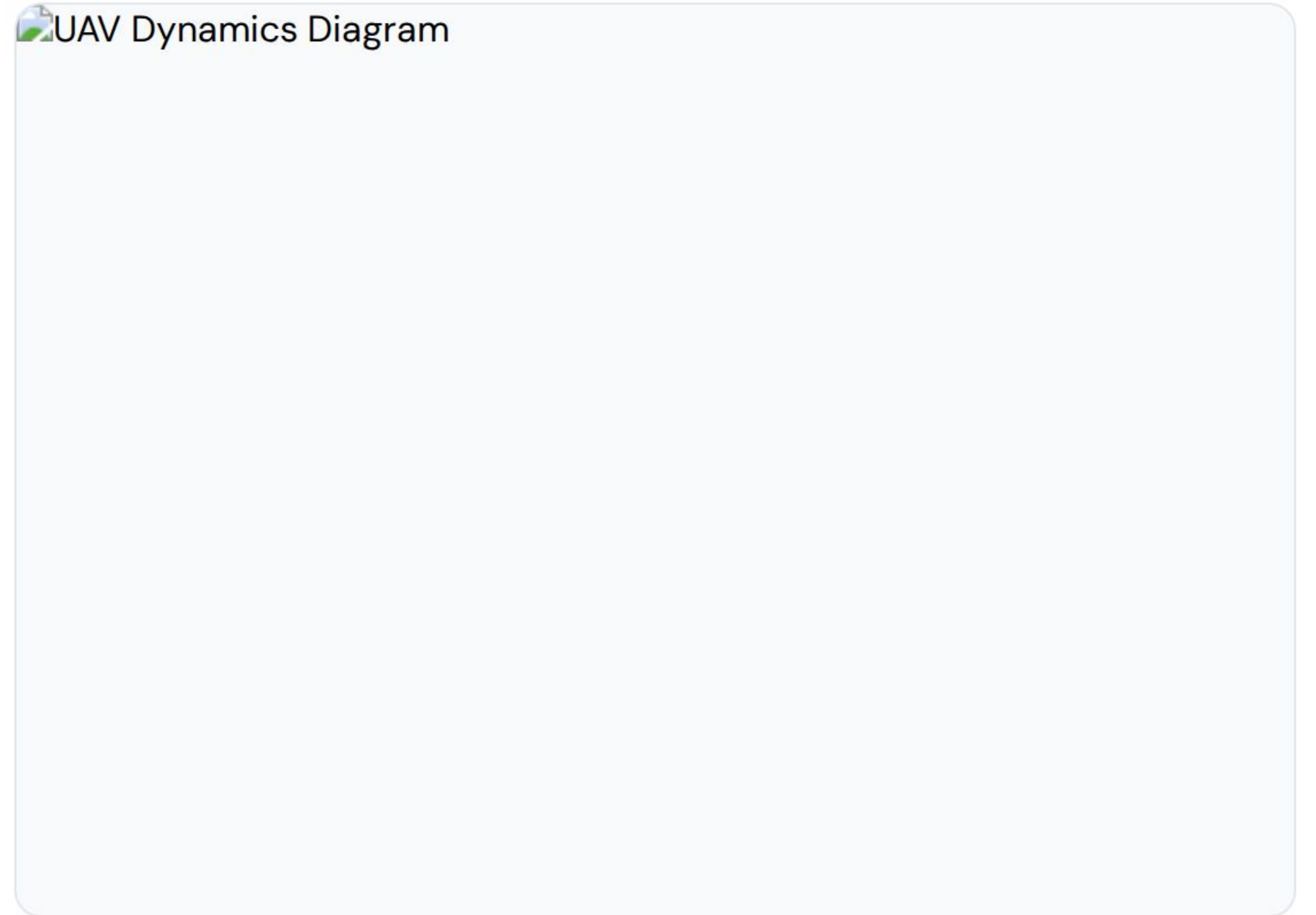
Underactuated Systems

Quadrotors have 6 degrees of freedom but only 4 independent control inputs (motor speeds).

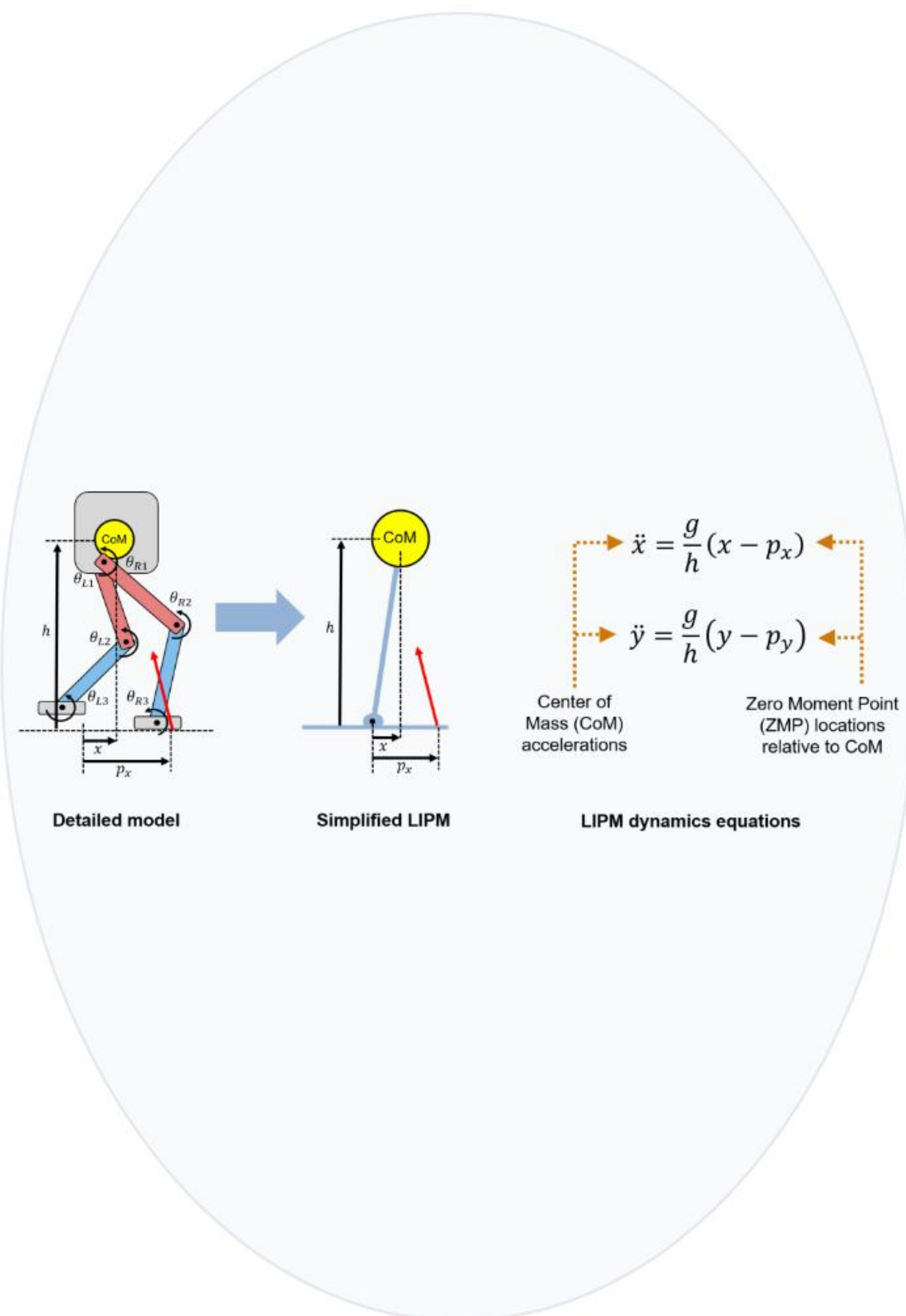
✈ Thrust for altitude.

🔄 Differentials for Roll, Pitch, and Yaw.

 UAV Dynamics Diagram



Humanoid Locomotion & Balance

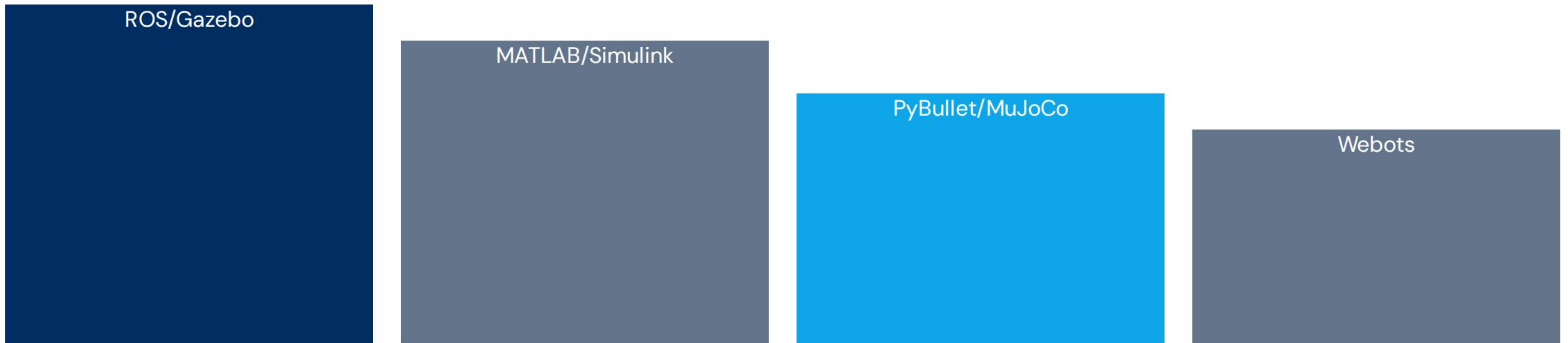


Zero Moment Point (ZMP)

For a biped to remain stable while walking, the ZMP must reside within the support polygon formed by the feet.

- 🚶 Linear Inverted Pendulum Model (LIPM).
- ⚙ Real-time Capture Point control for push recovery.
- ⚙ Whole-body control (WBC) for task prioritization.

| Simulation & Toolchains



Comparison of Industry Preference for Robotic Simulation Frameworks (Estimated usage in Research vs Production).

| Emerging Trends in Robotics



Reinforcement Learning

End-to-end learning for complex manipulation tasks where modeling is difficult.



Soft Robotics

Compliant actuators and structures for safe human interaction and delicate handling.



Swarm Control

Decentralized coordination of multiple agents for large-scale environmental monitoring.

| Thesis Summary & Takeaways

- ✔ **Integrated** Effective control requires a deep **Approach:** understanding of both mechanical dynamics and control theory.
- ✔ **Transition to** While PID is standard, advanced nonlinear **Nonlinearity:** methods are mandatory for agile and safe robots.
- ✔ **Simulation** Modern toolchains (ROS, MuJoCo) are **is Key:** indispensable for validating controllers before deployment.



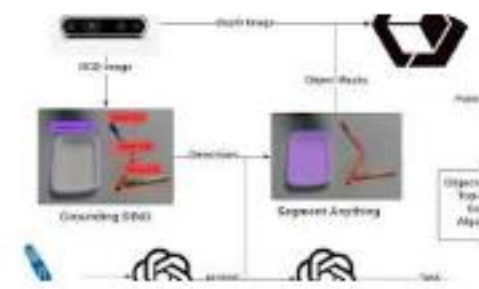
Q & A

Thank you for your attention.

Contact: researcher@university.edu

Thesis Repository: github.com/robotics-control-lab

| Image Sources



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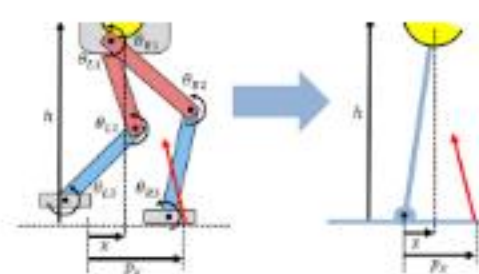
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